Swaraj Sharma Durgi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.61
NARAYANA E-TECHNO SCHOOL THANE MR	Class XII (Central Board of Secondary Education)	2024	94.80%
Narayana Olympiad School, Baghamberpet HYD	Class X (Telangana State Board of Secondary Education)	2022	9.00%

PROFESSIONAL EXPERIENCE

Calligo Technologies

Intern: Compute Acceleration & Hardware-Aware DL

- **SoC-Integrated FPGA-Based Posit GEMM Hardware Accelerator:** Engineered a parallelized, high-throughput GEMM hardware core for alternative numeric formats (Posit/Unum Type-III). Designed a parallel compute engine featuring format extraction, uncompressed dot-product accumulation, and re-encoding to isolate intermediate arithmetic from quantization noise. Architected ultra-wide parallel AXI4 memory-mapped data paths and control channels to move entire macro-matrix slices. Integrated an interrupt-driven DMA controller to offload multi-channel data movement from the host processor.
- **Alternative Numeric & Microscaling-based Quantization Frameworks:** Developed custom sub-16-bit block-scaled Posit quantization paradigms, demonstrating enhanced dot-product accuracy over standard microscaling baselines by exploiting the unique properties of Posits and distribution of data. Developed comprehensive PyTorch quantization simulation engines to evaluate numerical convergence, inference accuracy, and hardware resource bounds.
- **DL Inference Core:** Designed a fully pipelined, hardware-aware neural network inference core integrating linear operators and non-linear activation layers. Resolved deep physical routing paths and setup/hold violations by executing Static Timing Analysis to guarantee timing closure.

RESEARCH AND PUBLICATIONS

Function-Correcting Codes with Optimal Data Protection for Hamming Code Membership

***Swaraj Sharma Durgi**, Anjana A. Mahesh, Anupriya Kumari, Rajlaxmi Pandey, B. Sundar Rajan*

Proceedings of IEEE International Symposium on Information Theory (ISIT) 2026

- Pioneered the application of previously theoretical Function-Correcting Code (FCC) paradigms to algebraic code frameworks.
- Characterized the discrete geometric properties of the Hamming Code for the complex FCC Hamming Code Membership Function problem.
- Engineered a systematic parity-assignment construction exploiting the discrete geometry of distance-3 relations graphs within the classical [7,4,3]-Hamming code space.
- Proved mathematically that this tailored construction yields optimal data protection.
- Executed comprehensive AWGN channel simulations, demonstrating massive Bit Error Rate (BER) optimization over arbitrary valid Function-Correcting Codes.

PROJECTS

Multi-Core MESI Cache Coherence Simulator (Modern C++, Python) GitHub Repo

- Built an object-oriented C++17 simulator to model state transitions in private L1 direct-mapped caches operating under the MESI coherence protocol over a shared bus topology.
- Parsed multi-threaded memory access traces to track line state transitions, bus reads/invalidations (BusRd, BusRdX), and cache-to-cache data transfers.

- Automated Python test scripts to sweep core counts (2 to 8) and cache sizes (64 to 512 lines) to measure bus contention and invalidation overhead under shared versus isolated memory workloads.

Digital Signal Processing & Arithmetic RTL Design

- Designed an 8-point Radix-2 SDF FFT engine and a series of 101-tap FIR filter in Verilog running at 50 MHz on an Intel MAX 10 FPGA; some including folded symmetric FIR taps to reduce DSP multiplier utilization by 50%.
- Applied 5-stage pipelining, 3-stage complex multipliers, and adder trees to shorten critical paths and meet static timing constraints.
- Built a signed 32-bit Q16.16 Newton-Raphson divider with a custom Leading Zero Detector (LZD) and a 32-entry LUT for 3-cycle convergence, verifying outputs against Python golden models.

CUDA Kernel for Posit MatMult Sim GPU Parallelization (CUDA, C++, Python, PyTorch) GitHub Repo

- Built a C++/CUDA simulation engine for dynamic, variable-width Posit matrix multiplication (up to 32-bit width with configurable exponent sizes) using custom C++ template generation via CuPy.
- Wrote CUDA device functions implementing a 128-bit quire storage structure to execute exact fused multiply-accumulate arithmetic without intermediate rounding loss.
- Optimized global memory bandwidth using 16x16 shared-memory tiling, and integrated DLPack zero-copy tensor passing to exchange data directly between PyTorch and GPU device arrays without host copies.

GPU SIMT Control Flow Simulator GitHub Repo

- Built a C++ execution engine modeling GPU warp branch divergence and thread reconvergence based on Fung et. al's Immediate Post-Dominator (IPDOM) stack architecture.
- Implemented hardware reconvergence stack logic utilizing 32-bit active lane masks to track taken, fallthrough, and reconvergence execution paths across divergent threads.
- Measured SIMD lane utilization efficiency and cycle overhead across uniform, interleaved (checkerboard), and single-thread branch predicate patterns.

TECHNICAL TOOLKIT

Languages: Verilog, C++, Python, CUDA, C, Assembly (RISC-V, AVR, ARM), MATLAB, Julia

Hardware & EDA: FPGA Development (Vivado, Quartus, Libero), ModelSim/Questa, Icarus Verilog, GTKWave, cocotb, LTspice, gem5, Oscilloscopes, Raspberry Pi, Microcontrollers

Frameworks: PyTorch, NumPy, CuPy, DLPack, OpenCV, Git, Linux Operating System Internals, Alternative Numeric Simulators

Relevant Coursework: Microprocessor & Computer Arch, Machine Learning, Digital Systems, DSP Lab, Introduction to CMOS Processing, Analog Electronics, Signal Processing, Control Systems, Data Structures & Applications, Matrix Theory, Probability, Random Processes

CO-CURRICULAR ACTIVITIES

- **Core Member, Books and Movies Club (2025-2026):** Managed operations, collaborative team efforts, and regular events.
- **Creative & Performing Arts:** Active bassist and keyboardist in a band; hobbyist photographer, creative writer, and avid reader.